Exchange-Closed Geometry and the Pancake Control Program

Working status

This is an internal research note for the Emergent Horizon Framework / Relational Kinematics Atlas program. It records the pancake control probes and the perfect-matching positive-control probe that were run before moving into the QM bridge.

The central question was:

> Are Johnson graphs genuinely special for EHF/QM-style structure, or does any local reversal/permutation system produce similar force and spectral behaviour?

The answer from the current probes is:

> Johnson graphs are special, but not isolated. They appear to belong to a broader class of exchange-closed geometries. Perfect matchings also fall into this class. Pancake graphs are nearby and useful, but they do not have the same exact spectral closure.

Executive summary

We compared three related finite combinatorial substrates:

1. **Johnson graphs** — subset-exchange geometries.
2. **Perfect matching graphs under 4-flips** — exchange-closed matching geometries.
3. **Pancake/local reversal graphs** — permutation worlds generated by local reversals.

The tests focused on whether a natural observable gives an exact harmonic-restoring drift and/or exact spectral closure.

The core observable-eigenmode test is:

$$
P h_c = \lambda h_c,
$$

where:

- P is the random-walk operator on the graph,
- h is the candidate coarse observable,
- $h_c=h-\mathbb{E}[h]$ is the centered observable.

The result:

Substrate	Observable	Exact drift?	Exact spectral observable?	Interpretation
Johnson $J(2k-1,k)$	subset sum $h(S)=\sum_{i\in S}i$	yes	yes	positive control
Perfect matchings on $2m$ points	span sum $h(M)=\sum_{(a,b)\in M}\|a-b\|$	yes	yes	positive control
Pancake/local reversals on S_n	inversion number $h(\pi)$	near-linear only	no	semi-positive control
Pancakes with refined key	(h,d,τ)	exact for Δh law	no	local-transition atlas, not full spectral closure

This sharpens the EHF claim. The result is **not** that every local graph gives quantum-like structure. The result is more precise:

> Exact QM-relevant spectral closure appears in special exchange-closed geometries. Local reversal systems can produce force-like behaviour, but that alone is insufficient.

Background definitions

Johnson graph

A Johnson graph $J(N,k)$ has vertices given by k-element subsets of $[N]$. Two vertices are adjacent if the corresponding subsets differ by exactly one element.

In the even-side EHF/RKA construction, the relevant Johnson family is often:

$$
\Omega_k=\{S\subset[2k-1]: |S|=k\}.
$$

The shell observable is:

$$
h(S)=\sum_{i\in S}i.
$$

The shell centre is:

$$
h^*=k^2.
$$

A known exact drift law from the EHF/RKA force-atlas work is:

$$
A(h)=E[\Delta h\mid h]
=
\frac{2k-1}{k(k-1)}(k^2-h).
$$

\]

Equivalently, for the centered observable \(h_c=h-k^2\), the random-walk operator satisfies:

\[
P h_c = \lambda h_c.
\]

So the linear force law is also an exact spectral statement.

Pancake graphs and local reversal graphs

The standard pancake graph has vertices given by permutations of \(n\) objects. A move flips the top \(r\) entries of the permutation. In the usual pancake graph, allowed flips are prefix reversals of length \(2,3,\dots,n\).

For the present EHF control program, the main pancake-like world was the local contiguous-reversal graph:

\[
S_n \quad \text{with all contiguous reversals of length }2\text{ or }3.
\]

This differs from the classical prefix-only pancake graph, but it is more relevant to the local-reversal logic used in EHF/RKA.

The main observable was inversion number:

\[
h(\pi)=\#\{(i,j): i<j,\ \pi_i>\pi_j\}.
\]

The inversion-centre is:

\[
h^*=\frac{n(n-1)}{4}.
\]

Perfect matching graph under 4-flips

The perfect matching world has vertices given by perfect matchings on \(2m\) labelled points.

A local 4-flip chooses two matched pairs:

\[
(a,b),\ (c,d)
\]

and replaces them with one of the other two pairings among a,b,c,d.
 This gives a regular graph of perfect matchings.

The natural observable tested was total span:

$$
h(M)=\sum_{(a,b)\in M}|a-b|.
$$

The mean span is:

$$
h^*=\frac{m(2m+1)}{3}.
$$

Probe 05 found an exact restoring law:

$$
E[\Delta h\mid M]
=
-
\frac{2m-1}{m(m-1)}
\left(h(M)-h^*\right).
$$

Equivalently:

$$
P h_c=\lambda h_c.
$$

Probe log

Probe P01 — Pancake lumpability audit

Purpose

Establish whether the pancake/local reversal world behaves like Johnson
 shells under the inversion observable.

The initial test used:

$$
S_n \quad \text{with contiguous reversals of length }2,3.
$$

The main shell observable was inversion number h.

What was measured

- number of vertices,

- degree,
- connectedness,
- inversion drift linearity,
- strong shell lumpability by inversion shell.

Result

For $n=4$ through 8, the graph is connected and regular. The inversion drift is close to linear and centred at:

$$
h^*=\frac{n(n-1)}{4}.
$$

Observed drift linearity for local $L=2,3$:

n	vertices	degree	connected?	inversion drift R^2
4	24	5	yes	0.962958
5	120	7	yes	0.968668
6	720	9	yes	0.976369
7	5040	11	yes	0.982042
8	40320	13	yes	0.986095

However, inversion shells are not strongly lumpable.

For $n=7$, local $L=2,3$:

```text
h-shell strong mean TV = 0.110862
h-shell strong max TV  = 0.265879
```

This means two permutations with the same inversion number can still have different transition behaviour.

EHF verdict

This weakens the naive claim:

> Any local reversal graph automatically gives Johnson-style shell mechanics.

But it strengthens the sharper EHF claim:

> Johnson-style closure is special and must be earned by the structure of admissible comparison.

Probe P02 — Pancake missing-coordinate probe

Purpose

Find the hidden coordinate responsible for the pancake residuals.

The main question was:

> Why is the inversion drift almost linear but not exact?

Candidate refinements included:

- descents,
- major index,
- local reversal-window statistics,
- local triple statistics.

Key definitions

Adjacent descents:

$$
d(\pi)=\#\{i: \pi_i>\pi_{i+1}\}.
$$

Consecutive triple inversion total:

$$
t_3(\pi)=\sum_i \operatorname{inv}(\pi_i,\pi_{i+1},\pi_{i+2}).
$$

Exact formula found

For local $L=2,3$ reversals, the state-level inversion drift is
 controlled exactly by local order-pattern information:

$$
E[\Delta h\mid \pi]
=
\frac{(n-1)-2d+3(n-2)-2t_3}{2n-3}.
$$

The script verified this exactly:

```text
local23 exact formula check: max_abs_error=0.000e+00
```

Key numerical result at $n=7$

Key	First-moment RMS error	Strong lumpability mean TV
h	0.3070037	0.1108616
(h,d)	0.1501967	0.2819982
t_3	0.08963385	0.2830291
(d,t_3)	approximately 0	0.3620652
(h,d,t_3)	approximately 0	0.4384597

Interpretation

The pair \((d,t_3)\) closes the **force / first-moment drift** exactly, but it does not close the full transition law.

So pancakes have:

\[
\text{exact local force closure}
\]

but not:

\[
\text{Johnson-style reduced shell mechanics}.
\]

EHF verdict

This strengthens the atlas idea. Pancake inversion number is too coarse. The residual is not random noise; it encodes local pattern structure.

Probe P03 — Pancake local transition atlas

Purpose

Probe P02 closed the first moment. Probe P03 asked whether a refined local coordinate can close the entire \(\Delta h\) transition law.

Refined statistic

Define:

\[
\tau=(\tau_0,\tau_1,\tau_2,\tau_3),
\]

where \(\tau_j\) is the number of consecutive triples containing exactly \(j\) internal inversions.

The compact refined key is:

\[
(h,d,\tau).
\]

Main result

For local pancake reversals \(L=2,3\):

\[
\boxed{(h,d,\tau)\text{ gives exact inversion-transition closure.}}

\]

That means the law of \(\Delta h\) is determined by \((h,d,\tau)\).

However:

\[
\boxed{(h,d,\tau)\text{ does not give Johnson-style self-lumpability.}}
\]

So the refined coordinate closes the inversion-output law, but not the whole dynamics on the refined atlas.

Key numerical result at \(n=7\)

Key	First-moment RMS	\(\Delta h\) law TV	destination-\(h\) TV	self-key TV	compression
\(h\)	0.307004	0.110862	0.110862	0.110862	0.00437
\(h,d,t_3\)	approximately 0	0.077318	0.077318	0.438460	0.04643
\(h,d,\tau\)	approximately 0	approximately 0	approximately 0	0.462708	0.09524
\(h,d,t_3,\text{pat3}\)	approximately 0	approximately 0	approximately 0	0.523606	0.18968

Compact \(n=8\) full-transition check

Key	First-moment RMS	\(\Delta h\) law TV	destination-\(h\) TV	self-key TV
\(h\)	0.300691	0.104593	0.104593	0.104593
\(h,d,t_3\)	approximately 0	0.078649	0.078649	0.455063
\(h,d,\tau\)	approximately 0	approximately 0	approximately 0	0.527686

Interpretation

Pancakes have a real local-transition atlas. But that atlas does not collapse into a clean Johnson-like reduced graph.

This is a valuable distinction:

- **force closure**: yes, after refinement;
- **inversion transition closure**: yes, after refinement;
- **full self-lumpability**: no;
- **spectral closure**: tested next, no.

Probe P04 — Pancake spectral atlas

Purpose

The QM bridge cares about operators and eigenmodes. Probe P04 asked:

> Does the pancake refined atlas recover clean low eigenmodes, or does failure of self-lumpability persist spectrally?

The test compared:

$$
h
\quad \text{versus} \quad
(h,d,\tau).
$$

Main result

Johnson has exact spectral observable closure. Pancakes only approximate it.

For Johnson:

World	Size	h eigen residual
Johnson $J(5,3)$	10	2.19×10^{-16}
Johnson $J(7,4)$	35	1.41×10^{-16}
Johnson $J(9,5)$	126	1.49×10^{-16}
Johnson $J(11,6)$	462	1.74×10^{-16}

For pancakes:

World	vertices	h eigen residual
Pancake $n=4$	24	0.391297
Pancake $n=5$	120	0.202817
Pancake $n=6$	720	0.138409
Pancake $n=7$	5040	0.102960
Pancake $n=8$, observable-only	40320	0.080068

Pancakes become more oscillator-like as n grows, but they are not exact at finite n.

Refined atlas result

The refined key (h,d,τ) closes the Δh law, but it does not become a clean spectral sector.

Observed feature-subspace invariance residuals:

n	(h,d,τ) feature-subspace invariance residual
4	0.209121
5	0.302863
6	0.299375
7	0.245746
8	0.206496

Interpretation

This is the QM-relevant distinction:

$$
\boxed{\text{Johnson gives exact spectral closure. Pancakes do not.}}
$$

Pancakes are therefore not equivalent to Johnson graphs in the QM bridge.

EHF verdict

This strongly supports the view that Johnson-like QM structure is not generic reversal behaviour.

Probe P05 — Perfect matching spectral exchange control

Purpose

After pancakes failed exact spectral closure, we needed a second positive control to determine whether Johnson was a one-off or part of a class.

The natural test was the graph of perfect matchings under local 4-flips.

Observable

For a perfect matching M on $2m$ labelled points:

$$
h(M)=\sum_{(a,b)\in M}|a-b|.
$$

Main result

Perfect matchings are an exact positive control.

For the random-walk operator P:

$$
P\, h_c=\lambda\, h_c.
$$

The residual is numerical zero through $m=7$:

pairs m	points	vertices	degree	h-eigen residual	drift R^2
2	4	3	2	5.00×10^{-16}	1
3	6	15	6	6.04×10^{-16}	1
4	8	105	12	5.53×10^{-16}	1
5	10	945	20	7.51×10^{-16}	1
6	12	10,395	30	3.35×10^{-16}	1
7	14	135,135	42	4.76×10^{-16}	1

Exact law

The mean span is:

$$
h^*=\frac{m(2m+1)}{3}.
$$

The exact drift is:

$$
E[\Delta h\mid M]
=
-
\frac{2m-1}{m(m-1)}
\left(h(M)-h^*\right).
$$

This is a clean entropic-restoring force law.

Important nuance

Perfect matchings do not necessarily have full strong lumpability by h-shell alone.

Example at $m=6$:

```text
h-shell destination TV mean ≈ 0.1591
h-shell destination TV max  ≈ 0.3512
```

But the observable itself is exact:

$$
P h_c=\lambda h_c.
$$

This matters because the QM bridge cares more about operator/eigen-observable closure than naive shell lumpability.

EHF verdict

This strengthens EHF because it shows Johnson is not merely a one-off coincidence.

The candidate class is:

$$
\boxed{\text{exchange-closed geometries}.}
$$

Consolidated interpretation

1. Pancakes are not Johnson graphs in disguise

The pancake/local reversal world has force-like behaviour, but it fails exact spectral observable closure for inversion number.

Even after adding the refined local key (h,d,τ), the system does not produce a clean spectral sector.

Therefore:

$$
\boxed{\text{local reversals alone are not enough.}}
$$

This is important because EHF uses local moves extensively. The pancake test protects us from overclaiming.

---

2. Johnson and perfect matchings look like exchange-closed geometries

Johnson graphs and perfect matching 4-flip graphs share a deeper property:

there is a natural observable h such that:

$$
P\ h_c=\lambda\ h_c.
$$

Equivalently, the expected one-step change is exactly linear in displacement from equilibrium:

$$
E[\Delta h]= -\alpha(h-h^*).
$$

This is structurally analogous to a harmonic restoring force.

This does not yet prove quantum mechanics. But it gives a precise bridge target:

> Find the finite combinatorial exchange geometries whose natural centered observables are exact eigen-observables of the admissible comparison operator.

---

3. Exact shell lumpability is not the only relevant criterion

The probes clarified a subtle but important distinction.

There are at least four levels of closure:

1. **First-moment closure**

   $$
   E[\Delta h\mid \text{state}]
   $$

   is determined by a coarse key.

2. **Output-law closure**

   The full distribution of Δh is determined by a coarse key.

3. **Self-lumpability**

   The full transition distribution into coarse-key classes is determined by the current coarse key.

4. **Spectral observable closure**

   The centered observable satisfies:

   $$
   P h_c=\lambda h_c.
   $$

Pancakes reach levels 1 and 2 after refinement, but not level 4.

Johnson and perfect matchings reach level 4 exactly.

For the QM bridge, level 4 is especially important.

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4. Why this matters for the QM bridge

Quantum mechanics is fundamentally operator-based. If EHF is going to bridge toward QM, we should prioritize structures where:

- observables are closed under dynamics,
- low eigenmodes are geometrically meaningful,
- multi-particle lifts produce spectral sums,
- coarse observables behave like finite harmonic modes,
- exact closure survives admissible comparison.

Johnson and perfect matchings are better candidates than pancakes because they have exact observable–eigenmode closure.

Pancakes remain useful because they are a nearby control showing what fails when the geometry is not exchange-closed.

Working conjectures

Conjecture 1 — Exchange-closed observable eigenmode principle

Let G be an admissible comparison graph with random-walk operator P. A natural observable h gives QM-relevant harmonic behaviour when:

$$
P(h-\mathbb{E}h)=\lambda(h-\mathbb{E}h).
$$

Johnson and perfect matching 4-flip geometries satisfy this exactly.

Pancake/local reversal geometry does not satisfy this exactly for inversion number.

Conjecture 2 — Local force closure is weaker than spectral closure

Local reversal systems can produce exact force laws after adding local pattern coordinates, but this does not imply exact spectral closure.

Pancakes are the current example:

$$
(h,d,\tau)\quad \text{closes the }\Delta h\text{ law, but not the spectral sector.}
$$

Conjecture 3 — QM bridge should begin from exchange-closed geometries

For the next phase, the QM bridge should prioritize substrates that satisfy exact observable-eigenmode closure, especially:

- Johnson graphs,
- perfect matching 4-flip graphs,
- Hamilton-cycle sector quotients that reduce to Johnson-like exchange geometries,
- possible higher exchange geometries or association-scheme-like structures.

Pancakes should remain as a negative/semi-positive control.

EHF strength assessment

Overall, these probes **strengthen EHF**.

They strengthen it for the right reason: they make the claim narrower and more falsifiable.

Before pancakes, there was a risk of overclaiming:

> Local moves naturally produce harmonic/QM-like structure.

After pancakes and matchings, the better claim is:

> Special exchange-closed admissible-comparison geometries produce exact harmonic/eigen-observable structure. Local reversal geometries may approximate this or expose local force atlases, but they do not automatically produce exact QM-like closure.

This is more defensible.

Recommended next work

Proof 01 — Perfect matching 4-flip drift law

The numerical result from Probe P05 can be promoted to an exact theorem.

Theorem

Let M be a perfect matching on [2m] = {1, 2, ..., 2m}. Define the span observable:

```text
h(M) = Σ_(a,b) in M |a-b|.
```

Let one step of the Markov chain choose two matched edges uniformly and replace them by one of the two alternative pairings of their four endpoints, uniformly. Then:

```text
E[Δh | M]
= - ((2m - 1) / (m(m - 1))) · ( h(M) - m(2m + 1)/3 ).
```

Equivalently, if P is the random-walk operator and hc = h - h∗, *where:*

```text
h∗ = m(2m + 1)/3,
```

then:

```text
P hc = (1 - ((2m - 1)/(m(m - 1)))) hc.
```

So the centred span observable is an exact eigen-observable.

Proof

Fix two matched edges of M. Their four endpoints, in increasing order, are:

```text
x1 < x2 < x3 < x4.
```

There are exactly three perfect pairings of these four endpoints:

```text
A = (x1,x2), (x3,x4)
B = (x1,x3), (x2,x4)
C = (x1,x4), (x2,x3)
```

Let their local span sums be sA, sB, and sC. Their sum is the total of all six pairwise distances among the four endpoints:

```text
sA + sB + sC = Σ_(1 ≤ i < j ≤ 4) (xj − xi).
```

Call this four-endpoint total SQ.

If the current local pairing has span sQ, then a 4-flip replaces it by one of the other two pairings. Therefore the average new local span is:

```text
(SQ − sQ) / 2.
```

So the expected local change for this chosen pair of matched edges is:

```text
ΔQ = ((SQ − sQ) / 2) − sQ
   = (SQ − 3sQ) / 2.
```

Now sum over all unordered pairs Q of matched edges. There are:

```text
C(m,2) = m(m − 1)/2
```

choices.

First:

```text
Σ_Q sQ = (m − 1) h(M),
```

because each matched edge appears alongside each of the other m − 1 matched edges.

Next define the total pairwise distance over all 2m labelled points:

```text
T = Σ_(1 ≤ i < j ≤ 2m) (j − i).
```

This has the closed form:

```text
T = (2m)((2m)^2 − 1)/6
  = m(4m^2 − 1)/3.
```

When summing SQ over all choices Q:

- every non-matched vertex pair appears exactly once;
- every matched vertex pair appears m − 1 times.

So:

```text
Σ_Q SQ = T + (m − 2)h(M).
```

Therefore:

```text
Σ_Q ΔQ
= 1/2 · (T + (m − 2)h(M)) − 3/2 · (m − 1)h(M).
```

The h(M) coefficient simplifies as:

```text
(m − 2) − 3(m − 1) = 1 − 2m = −(2m − 1).
```

Hence:

```text
Σ_Q ΔQ = 1/2 · (T − (2m − 1)h(M)).
```

Divide by the number of unordered edge-pair choices, m(m − 1)/2:

```text
E[Δh | M] = (T − (2m − 1)h(M)) / (m(m − 1)).
```

Using:

```text
```

```
T = m(4m^2 − 1)/3
  = m(2m − 1)(2m + 1)/3,
```

we get:

```text
T / (2m − 1) = m(2m + 1)/3.
```

Therefore:

```text
E[Δh | M]
= − ((2m − 1)/(m(m − 1))) · (h(M) − m(2m + 1)/3).
```

This proves the drift law.

Since P h = h + E[Δh | M], the centred observable hc = h − h* *satisfies:*

```text
P hc = (1 − ((2m − 1)/(m(m − 1)))) hc.
```

So the span observable is an exact eigen-observable of the 4-flip random-walk operator. Its eigenvalue is:

```text
λ = 1 − ((2m − 1)/(m(m − 1)))
  = (m^2 − 3m + 1)/(m(m − 1)).
```

EHF status

This strengthens EHF. The perfect matching result is now proof-level rather than numerical. It confirms that Johnson is not an isolated accident: at least one other nontrivial exchange geometry has the same exact harmonic/eigen-observable structure.

Next Probe B — Unified exchange-closed theorem template

Try to write one theorem covering both:

1. Johnson subset exchange,
2. perfect matching 4-flips.

The theorem should identify the structural reason that linear observables become exact eigen-observables.

Candidate language:

- exchange generator,
- labelled ground set,
- natural linear/span observable,
- reversible regular graph,
- centered observable eigenmode.

Next Probe C — Two-particle / Fock-style lift

For Johnson and perfect matchings, test:

$$
P^{(2)}=P\otimes I+I\otimes P
$$

or the normalized free lift.

Check whether eigenvalues combine as:

$$
\lambda_i+\lambda_j.
$$

This is the direct bridge back to the Fock-like behaviour already observed in EHF/RKA.

Next Probe D — Pancake asymptotic scaling

Pancake inversion residuals decrease with n:

n	pancake h-eigen residual
4	0.391297
5	0.202817
6	0.138409
7	0.102960
8	0.080068

This suggests pancakes may become asymptotically oscillator-like.

A useful control question:

> Does the residual vanish as $n\to\infty$, and if so at what rate?

This would distinguish exact finite exchange closure from approximate thermodynamic closure.

Unified exchange-closed theorem template

This section compares the Johnson and perfect-matching cases side by side and extracts the shared theorem shape.

Purpose

The goal is to identify what Johnson graphs and perfect matching 4-flip graphs have in common, without overextending the claim to arbitrary local reversal systems such as pancakes.

The shared structure is:

```text
finite exchange geometry + natural centred observable + reversible exchange walk
→ exact eigen-observable
→ exact linear restoring drift.
```

This is the current working definition of an exchange-closed geometry.

Side-by-side comparison

Feature	Johnson geometry	Perfect matching geometry
State space	k-subsets of [2k − 1]	perfect matchings on [2m]
State object	S ⊂ [2k − 1], |S| = k	M = set of m disjoint pairs
Local move	remove one selected label and insert one unselected label	choose two matched edges and re-pair the four endpoints
Move type	1-for-1 exchange	2-edge 4-point exchange
Graph	Johnson graph J(2k − 1, k)	perfect matching 4-flip graph
Degree	k(k − 1)	m(m − 1)
Observable	h(S) = Σ_*i in S i* | *h(M) = Σ_*(a,b) in M |a−b|
Centre	h∗ = *k^2* | *h∗* = m(2m + 1)/3
Drift form	E[Δh | state] = −α(h − h∗) | *E[Δh | state] = −α(h − h∗)*
Drift coefficient	α = (2k − 1)/(k(k − 1))	α = (2m − 1)/(m(m − 1))
Spectral form	P hc = λ hc	P hc = λ hc
Eigenvalue	λ = 1 − (2k − 1)/(k(k − 1))	λ = 1 − (2m − 1)/(m(m − 1))
EHF role	positive control	positive control

Here hc means the centred observable:

```text
hc = h − h∗.
```

Common theorem shape

Template theorem — exchange-closed observable eigenmode

Let Ω be a finite state space equipped with a regular reversible exchange walk P. Let h: Ω → R be a natural additive or span-like observable. Suppose the exchange generator has the following averaging property:

```text
E[Δh | x] = −α(h(x) − h∗)
```

for constants α > 0 and h∗ *independent of x*.

Then the centred observable:

```text
hc(x) = h(x) − h∗
```

is an exact eigen-observable of P:

```text
P hc = (1 − α) hc.
```

Conversely, if hc is an exact eigen-observable of P with eigenvalue λ, then:

```text
E[Δh | x] = (λ − 1)hc(x),
```

so the drift is exactly linear and restoring when λ < 1.

This is the operator form of the finite harmonic restoring law.

Johnson instance

State space:

```text
Ω_J = {S ⊂ [2k − 1] : |S| = k}.
```

Move:

```text
choose a ∈ S and b ∉ S, then replace a by b.
```

Observable:

```text
h(S) = Σ_i in S i.
```

Centre:

```text
h* = k^2.
```

Exact drift:

```text
E[Δh | S]
= ((2k − 1)/(k(k − 1))) · (k^2 − h(S))
= −((2k − 1)/(k(k − 1))) · (h(S) − k^2).
```

Therefore:

```text
P hc = (1 − ((2k − 1)/(k(k − 1)))) hc.
```

This gives exact observable closure.

Perfect matching instance

State space:

```text
Ω_M = perfect matchings on [2m].
```

Move:

```text
choose two matched edges and replace them by one of the two alternative
 pairings of their four endpoints.
```

Observable:

```text
h(M) = Σ_(a,b) in M |a−b|.
```

Centre:

```text
h* = m(2m + 1)/3.
```

Exact drift:

```text
E[Δh | M]
= −((2m − 1)/(m(m − 1))) · (h(M) − m(2m + 1)/3).
```

```

Therefore:

```text
P hc = (1 − ((2m − 1)/(m(m − 1)))) hc.
```

This gives exact observable closure.

What pancakes showed by contrast

Pancakes/local reversals do not fit this template using inversion number alone.

For local contiguous reversals of length 2 and 3:

```text
h(π) = inversion number
```

has near-linear restoring drift, but not exact eigen-observable closure.

The missing first-moment information is local-pattern data:

```text
d = number of adjacent descents
τ = histogram of consecutive triple inversion counts
```

The refined key (h,d,τ) closes the Δh law, but it still does not produce a clean spectral sector:

```text
(h,d,τ) closes local inversion-transition output,
but not full exchange-closed operator dynamics.
```

So pancakes separate three ideas:

1. local force behaviour,
2. refined transition-output closure,
3. exact exchange-closed eigen-observable structure.

Johnson and perfect matchings reach level 3. Pancakes currently reach levels 1 and 2 only.

Candidate definition — exchange-closed geometry

A finite admissible-comparison geometry is exchange-closed with respect to an observable h when:
```

1. the local move is an exchange operation on a finite labelled structure;
2. the move graph is regular and reversible;
3. h is natural for the labelled structure, usually additive or span-like;
4. the centred observable hc = h − E[h] is invariant as a one-dimensional operator sector:

```text
P hc = λ hc.
```

Equivalently, h has exact linear restoring drift:

```text
E[Δh | x] = −(1 − λ)(h(x) − E[h]).
```

This definition includes Johnson and perfect matching 4-flip geometries. It excludes the pancake inversion system at finite n.

EHF interpretation

This gives a sharper route into the QM bridge.

The proposed bridge is no longer:

```text
local moves → quantum-like structure.
```

The better bridge is:

```text
admissible comparison → exchange-closed emergent geometry → exact eigen-observable → harmonic/oscillator-like mode → many-body lift.
```

This is much more restrictive and therefore more useful.

The role of the Hamilton-cycle substrate is then to explain why Johnson-like or matching-like exchange geometries emerge from admissible comparison in the first place.

Immediate next theorem target

The next proof target is to abstract the shared counting mechanism.

For Johnson, the key identity is that summing all allowed label exchanges produces a global label total minus a multiple of the current selected-label sum.

For perfect matchings, the key identity is that summing all 4-flips
 produces the global pairwise-distance total minus a multiple of the
 current matching span.

The shared counting skeleton is:

```text
sum over local exchange windows of proposed-new observable
= global constant + local multiplicity correction − current-observable
 multiple.
```

If this skeleton can be formalised, it becomes the first general theorem
 for exchange-closed EHF geometries.

Final summary

The pancake program did its job.

It showed that local reversal worlds can produce force-like behaviour, but
 they do not automatically produce Johnson-style spectral closure.

The perfect matching program then showed that Johnson is not isolated. It
 belongs to a broader class of exchange-closed geometries where natural
 observables become exact eigen-observables.

The current best EHF/QM bridge direction is therefore:

$$
\boxed{\text{Hamilton-cycle substrate} \rightarrow \text{exchange-closed
 emergent geometry} \rightarrow \text{operator/eigenmode/Fock-like
 structure}.}
$$

Pancakes remain valuable as the nearby control case:

$$
\boxed{\text{local force atlas without full spectral closure}.}
$$